

#### Fourth Frontier of War.

Space has been militarised since the second half of 20th century. In the past, India had been studying space for acquisition of knowledge and for other peaceful purposes, and there was no intension to weaponise space. However, recent advances by our bordering countries for acquiring offensive and destructive space assets have compelled India to develop space warfare assets. Yet, India remains firm on its stance of using the domain of space only for peaceful purposes and does not support any form of weaponisation of outer space. Initially, the objective of placing military assets in space was primarily to enable the terrestrial forces to operate more efficiently and effectively.

The United States had been at the forefront of exploiting space for enhancing military capability, followed by Russia and China. In 2020, the United States established the U.S. Space Force as an independent dedicated military service.

#### Capacity-building

India is progressively becoming self-reliant and capable of maintaining deterrence in space and providing support to the armed forces on land, sea, and air. With increasing reliance on space based systems, these very same space based assets have become potential targets. Hence it is necessary to protect them to discourage the adversary. Most of the counter space technologies, such as surveillance, signal jamming, and spoofing of signals, are covert, including anti-satellite (ASAT) technologies.

From the military standpoint, the Indian defence forces need an integrated battle management command, control, and communications (BMC3) architecture for tracking and identifying anti-satellite weapons, providing communication jammers and sensors to overpower space-crafts. To provide effective BMC3 a variety of dedicated sensors are



Fig. 2: Anti-satellite (ASAT) weapon system Arrow 3 (Source: <https://autojournalism.com>)

needed for rapid tasking, processing, exploiting, and disseminating information across various levels of the chain of command.

Unlike the major space powers—the USA, Russia, and China—whose space programmes have military origins, India's space programme started as a civil project for peaceful purposes. As a result, the country developed limited capacity and experience in the military dimension of space domain. Today, India is facing the challenge of creating space assets such as co-orbital weapon systems and offensive cyber capabilities as a credible minimum deterrence against any threat.

Operating in space domain is highly technology-driven, cost-intensive, and demands financial and organisational capacity. Also, there is the danger of falling into the arms race spiral in space with a wide array of new weapon systems becoming available to the adversaries. To encourage widespread innovation and tapping into the public as well as private sector, academics, and the R&D facilities, the government is making available the required resources, including financial support.

#### Space situational awareness

Outer space is becoming essential for scientific and economic development as well as for ensuring security of the country. Hence the space regulatory authorities, research

institutes, defence agencies, spacecraft operators, launch providers, and space insurance organisations need real-time knowledge and data on the mission-related debris, rocket bodies, fragmentation debris, and functional and non-functional space-crafts. The space situational awareness (SSA) refers to gathering this knowledge and information of near-space environments and providing data for space traffic management and anticipating collision between the space-based assets and space debris. The SSA aids in providing the preventive measures to evade a 'first strike' on space assets.

For tracking the debris and monitoring the active and passive satellites in orbit, the Indian Space Research Organisation (ISRO) had been dependent on the data made available by North America Aerospace Defence Command (NORAD). Not being a NORAD member, ISRO could not access the real-time data, resulting in lack of accuracy, precision, and delay in locating the space debris. ISRO has now set up the Directorate of Space Situational Awareness and Management (SSAM) to enable India's space programme becoming more sustainable, secured, and independent.

To carry out the activities related to SSAM, the Space Situational Awareness Control Centre (SSACC) has been established in Bengaluru. The other SSA capabilities include